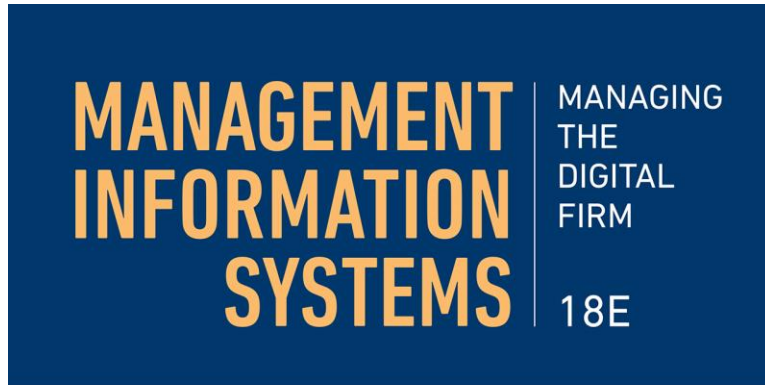


Management Information Systems: Managing the Digital Firm

Eighteenth Edition



Chapter 11

Artificial Intelligence



KENNETH C. LAUDON | JANE P. LAUDON | CAROL G. TRAVER

Learning Objectives

- 11.1** Understand artificial intelligence.
- 11.2** Describe machine learning.
- 11.3** Describe neural networks and deep learning.
- 11.4** Discuss generative AI.
- 11.5** Discuss intelligent agents.
- 11.6** Understand natural language processing.
- 11.7** Understand computer vision systems.
- 11.8** Describe robots.
- 11.9** Discuss genetic algorithms.
- 11.10** Discuss expert systems.
- 11.11** Describe fuzzy logic systems.
- 11.12** Understand the challenges of AI.
- 11.13** Understand how the information in this chapter can help your career.

Artificial Intelligence (1 of 2)

- Artificial intelligence (AI) systems have made great strides in their ability to perform tasks normally requiring human intelligence, such as
 - Speech recognition
 - Visual perception
 - Recognizing patterns
 - Making predictions
 - Writing essays
 - Learning from past experience

Artificial Intelligence (2 of 2)

- AI and human intelligence differ in their ability to adapt to new situations
 - AI systems are **designed to perform specific tasks** and are optimized for those tasks
 - Can learn to perform new tasks but require significant amount of training to do so
- AI systems have made significant strides in natural language processing (NLP) technology that gives computers the ability to interpret, manipulate, and comprehend human language
 - Still struggle to understand the **nuances** of human language and the **context** in which it is used

Major Types of AI Technologies

- Major branches of AI
 - Machine learning
 - Neural networks and deep learning
 - Genetic algorithms
 - Natural language processing
 - Computer vision systems
 - Robotics
 - Intelligent agents
 - Expert systems
 - Fuzzy logic

Machine Learning (1 of 2)

- Machine learning
 - Uses mathematical models that help a computer learn from large quantities of data and make statistical inferences without much explicit instruction
 - Enables a computer system to continue learning and improving on its own
 - Trained on large sets of data

Table 11.2**Examples of ML Applications**

Application	Description	Example
Natural language processing	Understanding and responding to human language	People can get personalized fashion advice tailored to their individual style preferences by conversing with a Stitch Fix chatbot.
Image recognition	Software that can identify objects, places, people, writing and actions in digital images.	Pinterest is an e-commerce visual discovery platform that allows users to visually share and discover new interests. It uses image recognition software for visual search. Users can choose specific items within a photo to find similar items. For example, selecting a throw pillow on a bed will return images of more throw pillows.
Predictive analytics	Predicting future events or trends based on historical data	Alibaba, the world's largest e-commerce platform, uses AI to predict what customers might want to buy.
Recommendation system	Recommending products or services to users based on their past behavior	Used by Amazon, Netflix, Spotify, Pandora, Apple Music, and many other companies to create personalized recommendations of products and services based on the preferences and choices of their customers.

Machine Learning (2 of 2)

- Supervised learning
 - System “trained” by providing examples of desired inputs and outputs identified by humans in advanced
 - One technique used to develop autonomous vehicles
- Unsupervised learning
 - Same procedures as used with supervised learning, but humans do not provide examples

Table 11.3

Examples of Unsupervised Learning Applications

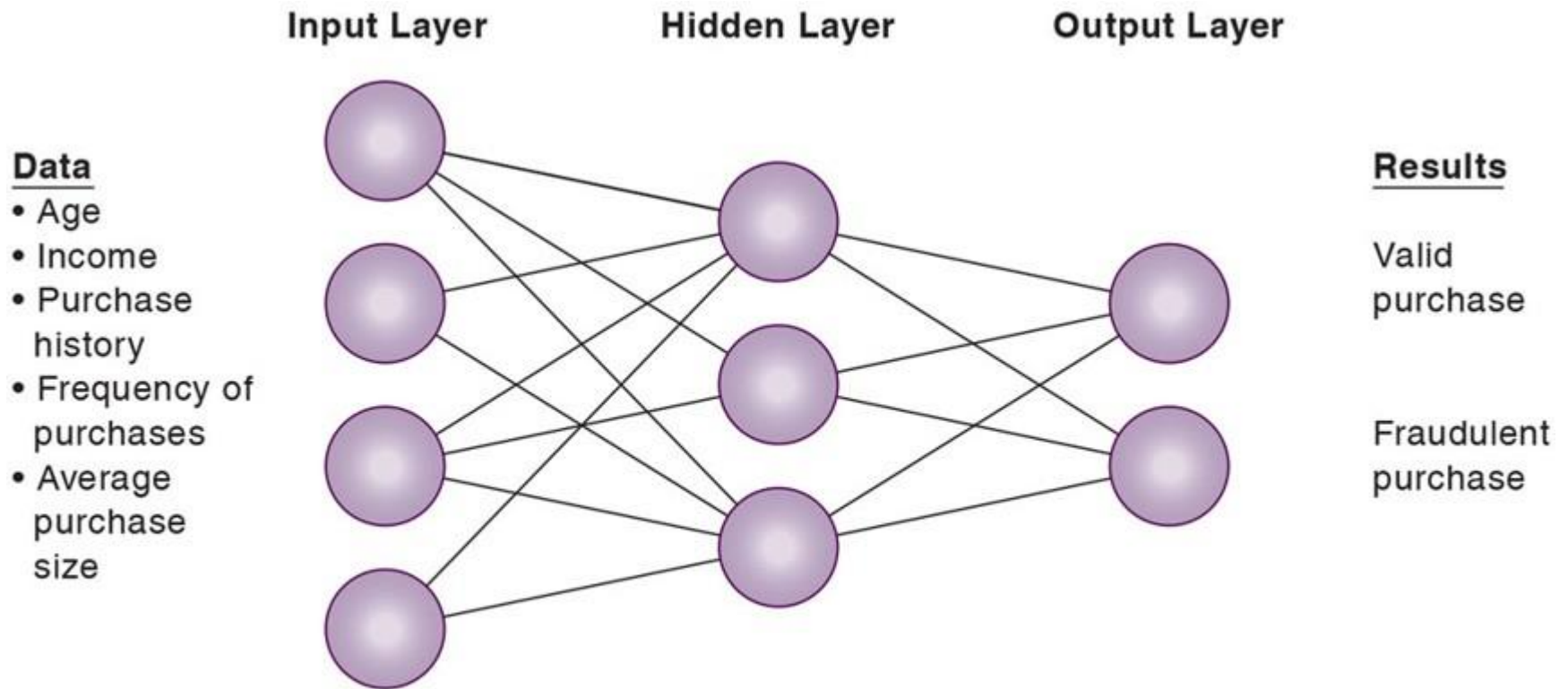
Application	Description	Example
Customer segmentation	AI segmentation tools group existing and potential customers based on shared characteristics and provide information for creating detailed customer profiles. This makes it possible to identify the right target audience and high-value customers.	Amazon uses unsupervised learning to segment customers based on their behavior and preferences to provide personalized product recommendations.
Medical imaging	Capabilities for medical devices, such as image detection, classification, and segmentation. Used in radiology and pathology to detect patterns quickly and accurately.	AltexSoft technology consultants created a prototype tool for chest X-ray analysis. The tool used machine learning to identify the boundaries of lungs and for lung disease classification.
Anomaly detection	Unsupervised learning can sift through large quantities of data and discover atypical data points within a dataset. These anomalies can call attention to abnormalities such as machine failures, product defects, or fraudulent credit card transactions.	JPMorgan Chase uses anomaly detection to track how customers typically use their credit cards. If a customer makes an abnormally large purchase or a purchase in a new location, the algorithm recognizes the anomaly and alerts a team member to contact the customer.
News aggregation	Gathering news information from various sources as it evolves on the web, analyzing it in real time, and organizing it into storylines.	Google News uses unsupervised learning to cluster articles on the same topic, such as World Cup soccer, from various online news outlets.

Neural Networks and Deep Learning

(1 of 3)

- Neural networks
 - Find patterns and relationships in massive amounts of data too complicated for humans to analyze
 - Learn patterns by searching for relationships, building models, and correcting over and over again
 - Humans “train” network by **feeding it data inputs for which outputs are known**, to help neural network learn solution by example from human experts
 - Used in medicine, science, and business for problems in pattern classification, prediction, financial analysis, and control and optimization

Figure 11.1 How a Neural Network Works



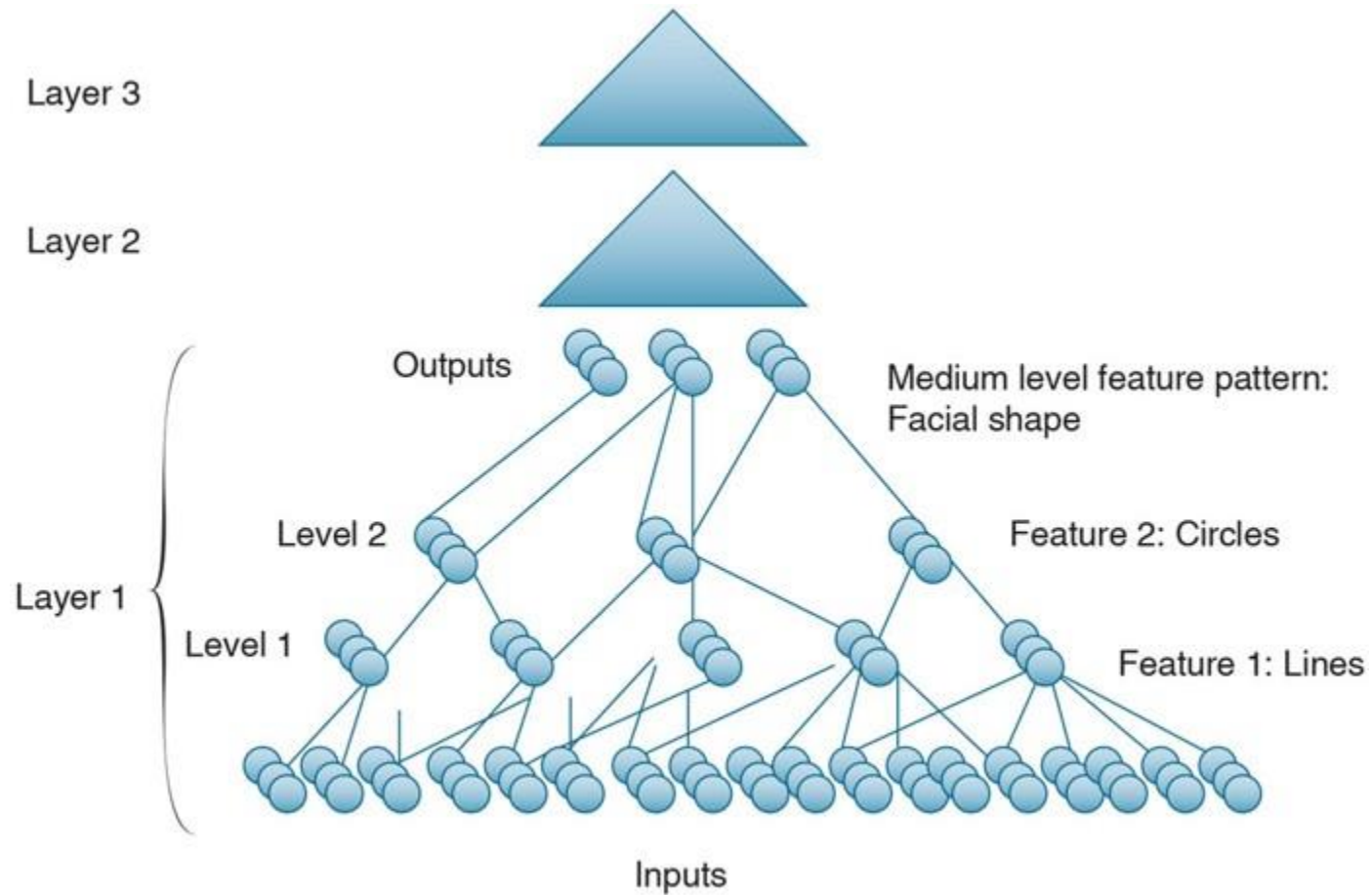
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Neural Networks and Deep Learning

(2 of 3)

- Deep learning neural network
 - More complex, **with many layers of transformations of input data to produce target output**
 - Used almost exclusively for pattern detection on unlabeled data (**unsupervised learning**)
 - Deep learning algorithms are powering many GenAI models

Figure 11.2 A Deep Learning Neural Network



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Neural Networks and Deep Learning

(3 of 3)

- Examples of neural networks and deep learning applications include
 - Computer vision
 - Speech recognition
 - Machine controls, diagnostics
 - Language translation
 - Transaction analysis
 - Targeted online ads

Table 11.4**Examples of Neural Networks and Deep Learning Applications**

Functionality	Inputs	Process	Outputs/Application
Computer vision	Millions of digital images, videos, or sensors	Recognize patterns in images, and objects	Photo tagging; facial recognition; autonomous vehicles
Speech recognition	Digital soundtracks, voices	Recognize patterns and meaning in soundtracks and speech	Digital assistants; chatbots; help centers
Machine controls, diagnostics	Internet of Things: thousands of sensors	Identify operational status, patterns of failure	Preventive maintenance; quality control
Language translation	Millions of sentences in various languages	Identify patterns in multiple languages	Translate sentences from one language to another
Transaction analysis	Millions of loan applications, stock trades, phone calls	Identify patterns in financial and other transactions	Fraud control; theft of services; stock market predictions
Targeted online ads	Millions of browser histories	Identify clusters of consumers; preferences	Programmatic advertising

Generative AI

- Generative AI (GenAI)
 - A machine learning model that is **trained to generate content**
 - Can identify patterns and structures within existing data to **produce new and original content**
 - GenAI starts with a prompt that could be in the form of text, an image, a video, a design, musical notes, or any input that the AI system can process
 - **Various AI algorithms then return new content in response to the prompt.**

Foundation Models, Large Language Models, and Transformers (1 of 3)

- Foundation model
 - Underlying model that enables GenAI to work
 - Uses powerful neural networks designed to mimic the human brain; “learns” by finding patterns in existing data sets
 - Includes models trained on other data types, such as images, audio, and video, or a combination of them (**multimodal**).
- Large language models (LLM)
 - a foundation model trained on immense amounts of unstructured data covering many topics
 - trained specifically on vast amounts of **text and code**

Foundation Models, Large Language Models, and Transformers (2 of 3)

- Transformer
 - A transformer is a key component of foundation models
 - For instance, GPT in ChatGPT stands for generative pre-trained transformer
 - Transformers are a type of neural network architecture that transforms or changes an input sequence into an output sequence
 - Include a capability called **attention** that enables models to track the connections between words across pages, chapters, and books rather than just in individual sentences.

Foundation Models, Large Language Models, and Transformers (3 of 3)

- GenAI tools
 - ChatGPT
 - Microsoft Copilot
 - Google Gemini
 - DALL-E

Intelligent Agents

- Intelligent agent
 - Software program that works in the background without direct human intervention
 - Carries out specific tasks for an individual user, business process, or software application
 - Smart thermostats
 - Driverless cars
 - Virtual assistants
 - Chatbots
 - Spam filters

Table 11.6

Types of Bots

Type of Bot	Description
Shopping bot	Virtual assistants that help shoppers during the buying and checkout process. Can also search the web for products and make price comparisons.
Web crawler	Search engine bot that scans and indexes web pages on the Internet; extracting data to understand the structure and relevance of web content.
Scraper	Web scraping crawlers, scan and download specific content on the internet. E-commerce businesses use scrapers to monitor live product prices on different retail platforms. Marketers use scrapers with natural language capabilities to run sentiment analysis on social media feeds
Monitoring bot	Constantly scanning systems for bugs and malicious software. They alert users to unusual web activity by collecting and analyzing user interaction data and web traffic.
Malware	Malicious bot that that can disable a system and steal data, or perform other harmful activities.

Natural Language Processing

- Natural language processing (NLP)
 - Makes it possible for a computer to analyze natural language— language that humans instinctively use
 - Can process voice or text command using natural human language
 - Examples: Google search; spam filtering systems; text mining sentiment analysis; customer call center interactions

Table 11.7

Examples of NLP Applications

NLP Application	Description
Conversational agents	NLP is the core technology behind AI chatbots and virtual assistants, such as Siri or Alexa. When you pose questions to these assistants, NLP is what enables them to not only understand the user's request, but to also respond in natural language. Conversational AI tools are frequently used for customer service in companies such as H&M, Domino's Pizza, and Bank of America to help customers find answers to common questions faster, reducing the load on customer service representatives.
Sentiment analysis	Sentiment analysis enables businesses to analyze customer sentiment towards brands, products, and services using online conversations or direct feedback. It is often used for analyzing social media content and conversations.
Text classification	By using NLP, text classifiers can automatically analyze text and then assign a set of pre-defined tags or categories based on its content. Once an algorithm is sufficiently trained on email textual data, it can successfully identify, categorize and label emails as regular emails, spam emails or malicious emails.
Automatic foreign language translation of text or speech	Tools such as Google Translate, DeepL, and Translate Me can translate text, audio and documents into another language. Netflix uses Google Translate to tailor content and subtitles to local audiences.
Text summarization	NLP tools can automatically read textual content and generate a summary. Text summarization informs users without requiring them to read every single detail, thus improving user productivity. The summary can be a news headline or a paragraph or several paragraphs of text. News aggregator websites and apps such as Apple News use NLP for text summarization.
Natural language content generation	NLP enables computers to generate text or speech that is natural-sounding and realistic enough to substitute for human communication. The generated language might be used to create drafts of reports, computer code, letters, memos or tweets. This requires on advanced language models trained on vast amounts of diverse data. These models, such as OpenAI's GPT, can generate coherent and contextually relevant sentences or paragraphs.

Computer Vision Systems

- Computer vision system
 - Uses machine learning and deep learning neural networks to emulate the human visual system to view and extract information from real-world images
 - Autonomous vehicles (drones and self-driving cars)
 - Industrial machine vision systems (e.g., inspecting bottles)
 - Military applications
 - Robotic tools

Table 11.8

Computer Vision System Applications

Computer Vision Application	Example
Facial recognition	Facial recognition uses computer vision to analyze a face in an image or video and compare it to a database of known faces to identify a person. Verkada security cameras with facial recognition capabilities allow businesses to identify authorized personnel and potential security threats.
Self-driving cars	Helps self-driving cars navigate their surroundings: Tesla relies on computer vision for its self-driving car technology, using cameras to detect objects and navigate the road.
Manufacturing fault detection	Inspects products on assembly lines for defects: Siemens utilizes computer vision for automated quality control, with systems inspecting products for defects on assembly lines.
Medical imaging	Analyze medical images, such as X-rays and MRIs, to help diagnose diseases: Stanford ML Group's CheXNet can detect pneumonia from chest X-rays with an accuracy exceeding that of practicing radiologists
Sports performance analysis	Identify diverse poses and actions of players to produce detailed position and action tracking models. SportVU 2.0 uses cameras and computer vision technology to capture detailed movement of both the ball in soccer or basketball and the players. Besides improving performance, football teams can use computer vision technology to prevent injuries.
Precision agriculture	Drones equipped with computer vision technology can identify specific areas infested with pests and diseases, allowing for targeted pesticide application. Ensures that only the necessary amount of chemicals is used, minimizing environmental impact and reducing costs. Acme Agritech offers drone and data analysis services to farmers, including drones equipped with specialized spraying mechanisms. These drones collected data to pinpoint problem areas and deliver precise amounts of fertilizer, pesticides, or herbicides only where needed.

Robotics (1 of 2)

- Robotics
 - Deals with the design, construction, and operation of movable machines to substitute for humans
 - Control, sensory feedback, and information processing
- Robotic applications
 - Commercially available applications include the use of Alto enable robots to sense and respond to their environment
 - Most widespread use of robotic technology has been in manufacturing and logistics

Robotics (2 of 2)

- Types of robots
 - Industrial robots
 - Collaborative robots (cobots)
 - Service robots
 - Humanoid robots
 - Medical robots
 - Drones
 - Autonomous cars

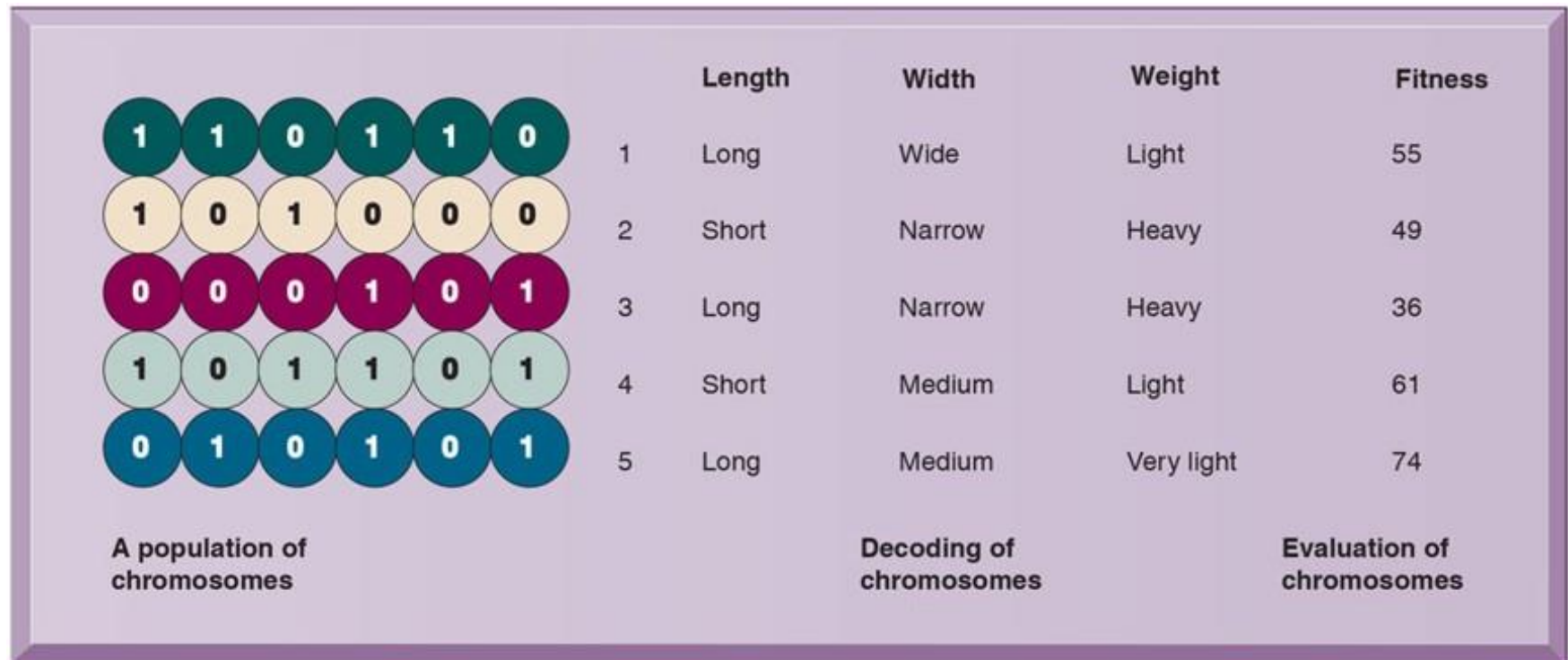
Sample Videos

- **Robots in Retail: Walmart robots at the Carlisle Pa. store**
 - <https://www.youtube.com/watch?v=MXuTtJkgmMk>
- **Miero - Robotik Eczane Sistemi Tanıtım videosu**
 - <https://www.youtube.com/watch?v=-90tA3-l5YU>

Genetic Algorithms

- Genetic algorithm
 - Another form of machine learning
 - Useful for finding the optimal solution for a specific problem
 - By examining a very large number of alternative solutions to that problem
 - Works by searching a population of randomly generated strings of binary digits to identify the string representing the best possible solution for the problem

Figure 11.3 The Components of a Genetic Algorithm



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Table 11.9

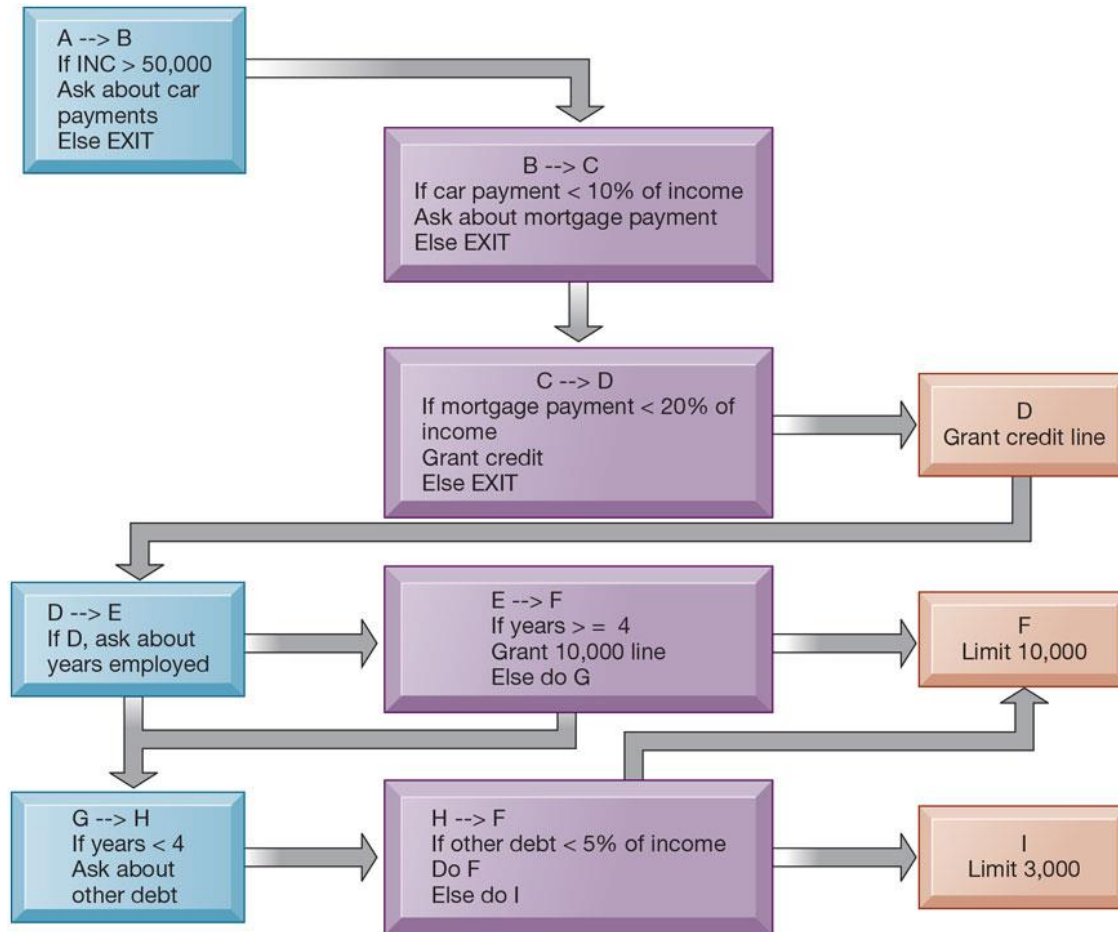
Examples of Genetic Algorithm Applications

Organization	Genetic Algorithm Application
Toyota	Used genetic algorithms to optimize production schedules, logistics routes, and inventory management, improving global supply chain efficiency and reducing costs.
Ford Motor Company	Used genetic algorithms to determine optimal routes for delivery vehicles, considering factors such as traffic conditions, package sizes, and delivery deadlines. Optimization helped streamline logistics operations, reduce delivery times, and improve overall efficiency.
Autodesk	Computer-aided design (CAD) and engineering software company incorporated genetic algorithms into its products. This enabled Autodesk software users to employ genetic algorithms for optimization problems, such as finding optimal shapes for mechanical components or generating efficient 3D structures.
NVIDIA	Used genetic algorithms to explore and fine-tune the design parameters of graphics processing units, improving performance and energy efficiency in AI and gaming applications.

Expert Systems

- Expert system
 - Captures the knowledge of individual experts in an organization and represents that knowledge as sets of rules
 - Often used when human expertise is critical and rules are well-understood

Figure 11.4 Rules in an Expert System

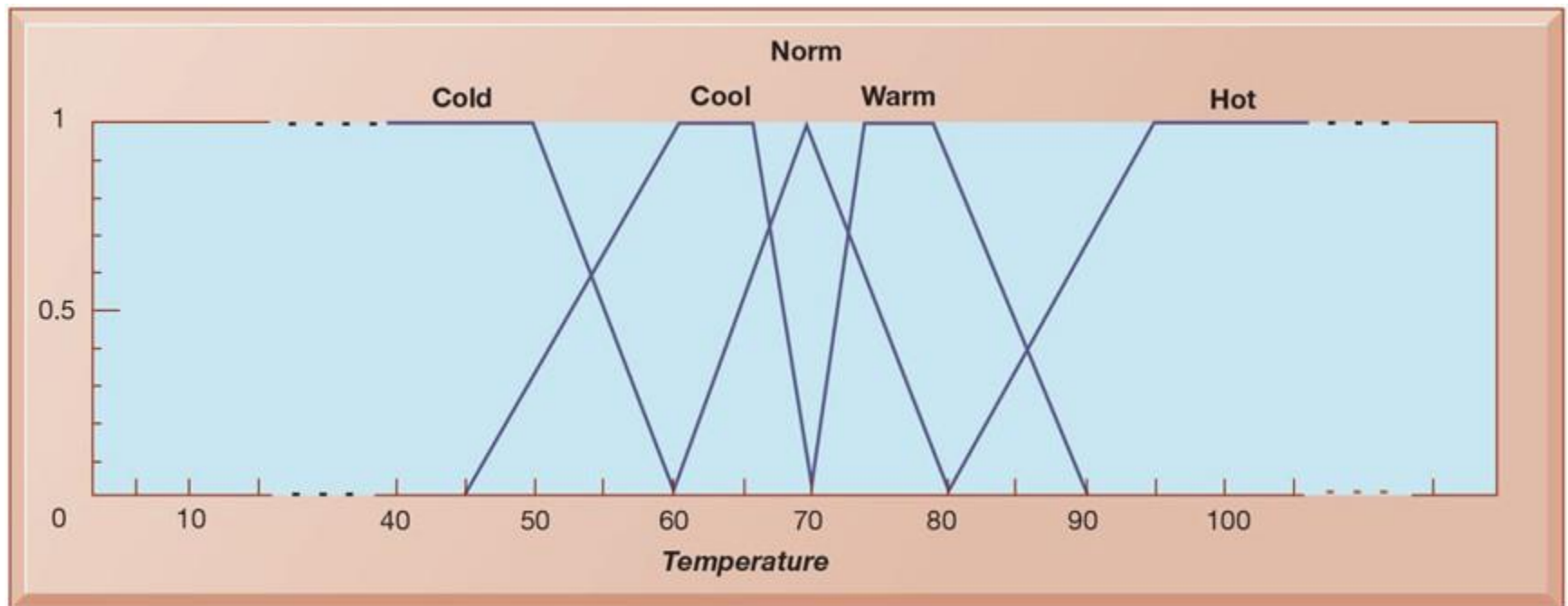


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Fuzzy Logic Systems

- Fuzzy logic
 - A form of AI that imitates human reasoning and cognition
 - Can describe a particular phenomenon or process linguistically and then represent that description in a small number of flexible rules

Figure 11.5 Fuzzy Logic for Temperature Control



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Table 11.10**Fuzzy Logic Applications**

Fuzzy logic application	Description
Automobiles	Fuzzy logic is used for gear selection and is based on factors such as engine load, road conditions and style of driving
Copy machines	Fuzzy logic is used to adjust drum voltage based on factors such as humidity, picture density and temperature
Environmental control systems	In air conditioners and heaters, fuzzy logic determines output based on factors such as current temperature and target temperature
Aerospace	Fuzzy logic is used to manage altitude control for satellites and spacecrafts based on environmental factors
Sendai subway system speed controls	Fuzzy logic controls make it possible for subway cars to accelerate very smoothly

Challenges of AI

(1 of 2)

- Hallucination
- Lack of transparency
- AI bias
- Cybersecurity problems
- Job loss and inequality

Challenges of AI

(2 of 2)

- Environmental impact
- Misinformation and disinformation
- Loss of privacy

Copyright and Intellectual Property Challenges

- The development of GenAI challenges the current copyright regimen for protecting intellectual property
 - A primary challenge is defining the level of human involvement required for AI-generated works to qualify for copyright protection
 - Web scraping
 - Often used by AI developers and vendors to “scrape,” or extract, content from publicly available sources on the Internet

Responsible AI

- Responsible AI is a set of practices to ensure that AI is designed, deployed, and used in an ethical and legal way
- These practices lead toward more beneficial and equitable outcomes

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